

Emoticontrol: Emotions-based Control of User-Interfaces Adaptations

MINA ALIPOUR, University of Southern Denmark, Denmark

MAHYAR T. MOGHADDAM, University of Southern Denmark, Denmark

KARTHIK VAIDHYANATHAN, IIIT Hyderabad, India

MIKKEL BAUN KJÆRGAARD, University of Southern Denmark, Denmark

Emotions are integral to human nature, and their existence, duration, and evolution could lead to specific behaviors. If emotions and behaviors are ignored in the design of socio-technical systems, they will fail or cause discomfort. User interfaces (UIs) are elements of interactive systems able to trigger or moderate emotions. UIs are increasingly designed adaptive to users' various characteristics, intending to improve their satisfaction, performance, and decisions. However, previous adaptation supervising approaches are not effectively adopted in real life since they neglect the dynamic behaviors of humans or systems. This paper proposes *Emoticontrol*, a quality-driven approach to adapting UIs to users' emotions using Model-Free Reinforcement Learning (MFRL). The approach aims to maximize applying the essential adaptations and minimize the unnecessary ones towards users' enhanced quality of experience (QoE). The approach also considers improving the software quality of service (QoS) by designing software architecture alternatives. We chose emergency evacuation training as a suitable evaluation domain since people experience intense emotions in potential danger. We performed experiments with a mobile application we developed that acts as a recommender system in evacuation training. By taking contextual input of the users' basic emotions from face recognition, the application intelligently adapts its UI to quickly lead people to safe areas while keeping them emotionally controlled. We consider software performance a crucial QoS; thus, we adopt and test architectures that facilitate an acceptable level of performance. The evaluation process confirms the efficiency and effectiveness of the MFRL in iterations, as well as compared to other UI adaptation techniques.

CCS Concepts: • **Human-centered computing** → **User interface management systems**.

Additional Key Words and Phrases: Emotions, User Interface, Self-adaptation, Reinforcement Learning, Emergency.

ACM Reference Format:

Mina Alipour, Mahyar T. Moghaddam, Karthik Vaidhyanathan, and Mikkel Baun Kjærgaard. 2023. *Emoticontrol: Emotions-based Control of User-Interfaces Adaptations*. *Proc. ACM Hum.-Comput. Interact.* 7, EICS, Article 175 (June 2023), 29 pages. <https://doi.org/10.1145/3593227>

1 INTRODUCTION

The advancements in connected socio-technical systems that enable perceiving and acting on human behaviors require establishing novel adaptation mechanisms. The mechanisms should link the interactive system to human characteristics and provide a desirable adjustment between the

Authors' addresses: Mina Alipour, mial@mmmi.sdu.dk, University of Southern Denmark, Odense 5230, Denmark; Mahyar T. Moghaddam, mtmo@mmmi.sdu.dk, University of Southern Denmark, Odense 5230, Denmark; Karthik Vaidhyanathan, karthik.vaidhyanathan@iiit.ac.in, IIIT Hyderabad, Hyderabad 500032, India; Mikkel Baun Kjærgaard, mbkj@mmmi.sdu.dk, University of Southern Denmark, Odense 5230, Denmark.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

2573-0142/2023/6-ART175 \$15.00

<https://doi.org/10.1145/3593227>

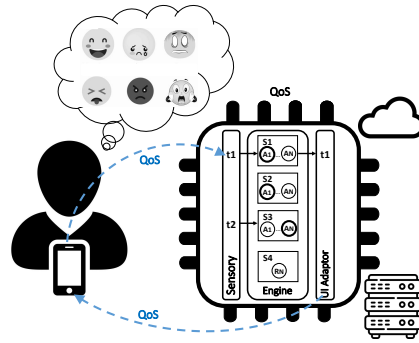


Fig. 1. MFRL for UI adaptation to emotions.

human quality of experience (QoE) and system quality of service (QoS). Some initial questions could be: are current interactive systems able to intelligently adapt to their own and the environment dynamics [1, 50]? Can those systems observe and react to what users tacitly want to see on their screens [68] or a manual interaction is needed [79]? Are the systems' performance compatible with real-time applications [13]?

Emotions can adapt interactive systems and their user interfaces (UIs). Emotions are a primary part of the human essence, leading to specific behaviors and impacting well-being. Thus, understanding emotions and considering them at real-time interaction is crucial for leading the users to specific behaviors [46]. Some researchers [17, 67] introduced UI self-adaptation approaches, such as model-based [71], rule-based [3], or optimization-based [38]. The difficulty is that human behavior can hardly be modeled, optimized, or structured within the rules [7, 18]. Humans may show fluctuating, unpredictable and irrational behaviors due to their individual, social, or environmental contexts [46]. Another issue is that modeling complex behaviors imposes system QoS degradation that could, e.g., delay systems' reaction to dynamic behaviors. Lack of consideration for those human and system dynamics can lead the interactive systems to failure, low quality, or discomfort for people.

To fill this gap, we propose *Emoticontrol*, a Model-Free Reinforcement Learning (MFRL) approach for UI adaptations to emotions. MFRL is a trial-and-error (learning from the environment by interacting with it) approach where the agent has to learn by taking actions from states repeatedly, which is known best for dynamic human behavior [64]. The type of MFRL used in this paper is called Q-learning [37]. This type of MFRL can be used when the state space and the action space are finite, meaning all possible states and actions are known. Q-learning maps all possible actions to all possible states in a Q-table, known as state-action pairs. The agent will take actions that result in changes to the state. The new state is checked against a reward table when an action is taken. The state maps to either a positive or a negative value in the reward table. The agent uses the reward/punishment value to calculate a value for taking that specific action from that specific state, aka. a value for that state-action pair. The value for the state-action pair is stored in Q-table as a value called the Q-value. Through trial and error, the agent will keep updating the Q-values in the Q-table. In a high-quality Q-table, the table Q-values represent a good or even the best action to take from a particular state.

Figure 1 shows how we adopt the MFRL approach as a supervisor for emotion-based UI adaptation. The approach is built around the *MAPEK* control loop [74], which includes *monitoring* raw emotion data, inferring and *analyzing* emotion values, *planning* and decision-making for adaptation to *execute*, with a global *knowledge-base*. In *Emoticontrol*, the knowledge-base that feeds the other run-time MAPEK elements receives input from design-time activities, i.e., emotion models, temporal

aspects, design parameters, baseline Q-Tables, and architectural configurations [51, 52]. In such systems, there are QoS concerns in perception, data transmission, computation, and actuation. The architectures could improve the QoS by modifying software components and their connection.

As shown in Figure 1, the built-in camera on a device detects and infers six types of Ekman basic emotions (namely joy, sadness, surprise, fear, disgust, and anger [20] plus neutral) that are *one of the state attributes in MFRL* and sends them to MFRL adaptation engine. The engine receives the emotional states each time and assigns reward/punishment values. Based on the values, a specific UI adaptation occurs (*an action in MFRL*) among various UI configurations (*the other state attribute in MFRL*) to possibly reduce the emotions associated with punishment and improve the emotions associated with reward. Q-Table values correspond to selecting an interface (*action*) from the set of interfaces (*state*).

As a **case study**, we apply the approach to a mobile application called *EvacuationApp*, developed for emergency evacuation training, to teach people how to safely and quickly reach a safe zone in emergencies. This training app is a suitable case since it pushes people to show intensive emotions in simulated danger [30] while avoiding real risks to human safety. We first surveyed 50 participants to collect insights and then implemented and evaluated the *EvacuationApp* in two iterations (with 24 and 27 participants performing 258 experiments). The application is designed to be used both indoors and outdoors. Training sessions using recommender systems like *EvacuationApp* could improve people's awareness and decision-making under risk. Using *EvacuationApp*, citizens place themselves in a hazard situation and follow an evacuation route to reach a safe zone. The users' location and their facial emotion data are continuously and anonymously tracked. The location information is used to give contextual awareness to people about their run-time position and surrounding environment. The emotion data is used to change the UI to impact users' emotions and consequently their mobility behavior to improve QoE.

This research aims at answering the following questions:

- **RQ1.** What UI parameters are associated with specific emotions?
- **RQ2.** How effective MFRL-based adaptation supervisor performs compared with other traditional approaches (i.e., rule-based, sequential, no adaptation)?
 - RQ2.1.** Do different adaptation methods arouse the target emotions?
 - RQ2.2.** What is the task completion rate associated with each adaptation method?
 - RQ2.3.** What is the correlation between target emotions and task completion?
- **RQ3.** Is the system QoS in line with the users' requirements?

With a literature review and survey, we provide scientific and experimental baseline, and further we implement our adaptive system and perform experiments to answer these research questions. The main contributions this paper makes are:

- A survey performed to understand the emotional consequences of UIs that could be re-used for similar research;
- The novel *Emoticontrol* approach and its adaptation engine developed for the community to be used in all interactive systems with users' emotions as a primary concern;
- An architecture approach that could enhance QoE of users and QoS of interactive software systems;
- Application of the approach to a real case.

The remainder of this paper is organized as follows. Section 2 presents a literature review. The research method is detailed in Section 3. Section 4 presents the *Emoticontrol* approach and how it is built at both design-time and run-time. Section 5 presents the survey results. Section 6 discusses the implementation of various pieces of the emergency training application and its adaptation engines, and Section 7 reports the results of conducted experiments. Section 8 presents the lessons learned,

and Section 9 explains the threats to validity. Section 10 concludes the paper and outlines future work directions.

2 STATE-OF-THE-ART REVIEW

User interface adaptation to emotions is interdisciplinary and requires unifying social, cognitive, and computer sciences. Firmenich et al. [22] discuss the two main techniques for *UI adaptations* in HCI that are known as tailor-made and rule-based adaptations. Tailor-made systems focus on defining adaptation at design-time or the beginning of a user session so that users can directly select or configure the desired version. Rule-based adaptation is instead based on programming the rules at design-time and updating a part or the whole UI at run-time based on dynamic contextual events. For instance, Zheng et al. [80] proposed a rule-based approach to adapt a user interface to various contexts, such as the user's age, color preferences, and weather status. Märtin et al. [41] developed a UI adaptation environment to improve the user's satisfaction while interacting with a system. Galindo et al. [23] developed a rule-based UI adaptation engine that can be integral to any web technology.

Apart from the methods mentioned above, machine learning methods [45] such as *reinforcement learning (RL)* could be adopted as an adaptation supervisor. Reinforcement learning focuses on learning what to do and how to map situations into actions to maximize a numerical reward signal. It divides the problem into four components: a policy, a reward function, a value function, and an internal environment model. There exist two main approaches in reinforcement learning: model-based and model-free. Model-based methods rely on planning as their primary component, while model-free methods primarily rely on learning. In a recent paper, Todi et al. [71] present an adaptive UI approach using model-based reinforcement learning to improve user satisfaction and avoid surprises. The approach was applied to an adaptive menu to enhance the menu organization for the user. However, the difficulty is that in the model-based approach, a model for each individual should be created. The agent would generate acceptable results in adapting to only the human for which the model is created. Yechiam et al. [78], Maia et al. [40], and Gläscher et al. [26] state that MFRL and its model-free approach of using Q-learning allow enacting individual level fine grained adaptations; however, it is not yet discovered in the emotion-based UI adaptation domain. Based on Kool et al. research [36], considering system performance, model-free strategies are computationally cheaper than model-based approaches.

Woodside et al. [77] define *Performance* as a pervasive quality affected by aspects of design, code, and execution environment. With their software performance engineering, Smith et al. [66] propose methods, such as measurement-based [70] and model-based [6] to evaluate and improve performance. While the measurement-based approach deals with testing, diagnosis, and tuning, the model-based approach adjusts the architecture to meet performance requirements. Based on Arbib et al. [4], the architecture specifies if computation resources should locate in the device, at the network edge, on the cloud, or a mix of them. Cawthon et al. [14], and Guynes [28] show that performance indices such as response time are dimensions of usability and impact emotions regarding tension, nervousness, apprehension, and fear.

The UI adaptation techniques mentioned above could be ideally triggered by users' detected **emotions**. Kleinginna et al. [34] define emotions as reactions to events and are tied up with the physiological, affective, behavioral, and cognitive state of a person. For instance, fear is an emotional reaction to a (negative) threatening situation [2, 65], and joy is a reaction to (positive) goal fulfillment. Brosch et al. [11] emphasize that emotions could be driven by external events called stimuli, like nice weather or a dark-themed UI. As emotions involve changes in our biological system, they can be detected from various measurements of biological changes [24] (e.g., heart rate or muscles' electrical changes) but also by the impacts of these changes on exposed behaviors, for

instance, by measuring changes in the face [19], gesture [27], or voice [62]. This research considers facial expressions as the source of emotions. Based on Konar et al. [35], facial expression analysis is the most common modality of emotion recognition.

There are two main theoretical categories in implying facial emotions: *the categorical models* [21] and *dimensional models* [53]. Categorical theories define emotions using tags from identified collections of emotions. The most prominent approach in this category is Ekman's basic (universal) emotions [20]. It represents six basic emotions of *anger*, *disgust*, *fear*, *joy*, *sadness* and *surprise*, which cover positive and negative emotions. Based on Savran et al. [61], and Meudt et al. [44], such emotions are defined as universal facial expressions and are generally stable over races, cultures, and ages and are consistent even in people blind by birth. [19] defined facial action coding system which measures facial expressions by identifying independent motions of the face in a group of action units that are the fundamental actions of one's muscles or muscles' group [12]. The dimensional model expresses emotions in a two or more-dimensional form. With the Circumplex Model of Affect theory, a theory in the dimensional category, Penser et al. [53] suggest that emotions have at least two dimensions in a circular space. These two dimensions are valence, which shows pleasure or displeasure (positive emotions and negative emotions), and arousal, which defines activation or deactivation (intensity of emotion) [75]. Bradley et al. [9] discuss other dimensional theories called self-assessment Manikin and Pleasure-Arousal-Dominance (PAD), which consist of valence, arousal, and dominance space. In this research, we adopt the Ekman basic emotions model since it is universal, well-established, and stable over various categories of users. This research proposes a suitable method for UI adaptation supervision, and while we study, capture, and analyze emotions, the precision of gathered emotion is not a key concern. It is also important to note that while basic emotions are expressed by all individuals regardless of culture or upbringing, the experience that produces them can be highly subjective. The consequence is that the emotions each individual feels are different. MFRL-based adaptation approach could tackle this challenge by providing a personalized experience for individuals.

This literature review studied UI adaptation, RL as adaptation supervisor, systems' performance dimensions, and emotions' importance in UI adaptation. Unifying and harmonizing these areas are crucial for understanding UI adaptation to emotions. Apart from understanding the gaps, we take advantage of the knowledge we gathered on emotion models, MFRL, and alternative adaptation approaches to concretely establish *Emoticontrol* and its design-time and run-time aspects.

3 RESEARCH METHOD

There are various methods to design and implement research. Researchers must make well-founded and informed decisions regarding research methodologies, techniques, and approaches in a given situation. Performing empirical research at the intersection of software engineering and HCI is challenging. The reason is that it requires studying the technology in use and social and cognitive processes. In addition, the implications of using individual methods are not well recorded. We structure this research by considering Wohlin's (2015) *software engineering research design method* [76] integrated with Pressman's (2019) *iterative approach* [54] and MacKenzie's (2012) *guidelines in scientific foundations of HCI* [39].

Wohlin [76] suggests following decision points, starting from identifying the problem and forming the research questions, and continuing with steps such as specifying expected outcomes, logic and purpose, approach, process, methodology, data collection and analysis, and finally findings. Each of these decision points is structured into constructive elements. We merged Wohlin's [76] guidelines with Pressman's [54] approach, where he suggests an iterative process, especially for implementation and experimentation. In his *Software Engineering: A Practitioner's Approach* [54], Pressman emphasizes that the stakeholders should be included early and often in the development

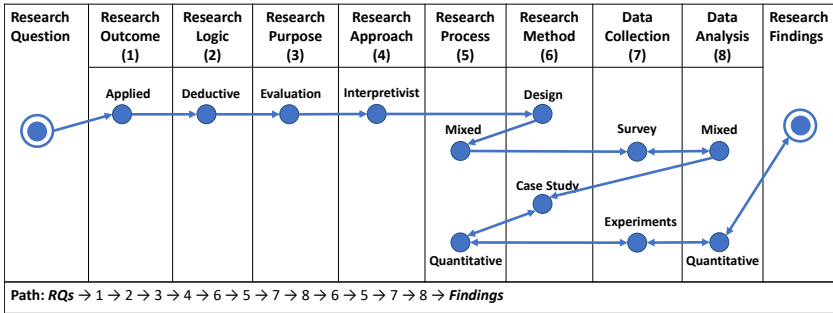


Fig. 2. The research methodology. Unidirectional arrows show the consecutive approach, and bidirectional arrows show the iterative approach.

process. The process starts with understanding the requirements of the users and specifying the focus. It follows with designing the first sketch of the system and deciding the methods and resources. Then the case goes through various iterations to achieve the functional and non-functional goals.

MacKenzie [39] defines the scientific foundations and practical guidelines of research papers in HCI. He introduces three common approaches, or methods, for conducting research in HCI: the observational method, the experimental method, and the correlational method. In the observational method, it is essential to observe the interaction. There are several observational methods (such as interviews and field studies) to discover and explain the reasons underlying human behavior. The experimental method acquires knowledge through controlled experiments conducted in laboratory settings. The correlational involves looking for relationships between variables. Our research gathers data through surveys and experimental measurements complemented by a correlational method for data analysis. MacKenzie [39] provides guidelines on designing experiments in HCI, hypotheses testing, and structuring scientific papers that we carefully considered in this research.

Figure 2 shows the methodology we followed. We defined the knowledge gap and formulated the research questions based on the literature review and our experience as human-computer interaction researchers, human behavior modelers, and software engineers, collaborating with industrial practitioners and governmental decision-makers. The *first* step in the figure specifies that we follow applied research to provide a better solution for UI adaptation and customization using MFRL. The *second* step shows how we adopt a top-down approach by forming, e.g., hypotheses and then performing experiments to prove or reject the hypotheses. The *third* decision point shows that we use evaluation by testing the effectiveness of MFRL compared with other similar methods to improve already existing systems. The *fourth* decision point specifies the interpretivist approach we adopted to understand human activities in specific contexts and situations. This approach is chosen since we believe that in emotion-based adaptation, the research is quite subjective to the human context.

Our methodology continues to the *sixth* decision point, considering the survey we conducted (see Section 5) to get insights from potential users into design, development, and emotions' consideration. For this step, we followed design science research addressed through building and evaluating artifacts designed to understand and further meet the requirements of a problem. To analyze the survey results, we considered a mixed research approach (the *fifth* decision point) to plan for both qualitative and quantitative research and data. The *seventh* decision point emphasizes that we used a survey for data collection. While surveys are generally considered a quantitative research method, we used both exploratory and descriptive ways to receive suggestions from users. The *eighth* decision point shows that we continuously analyzed the data using statistical methods

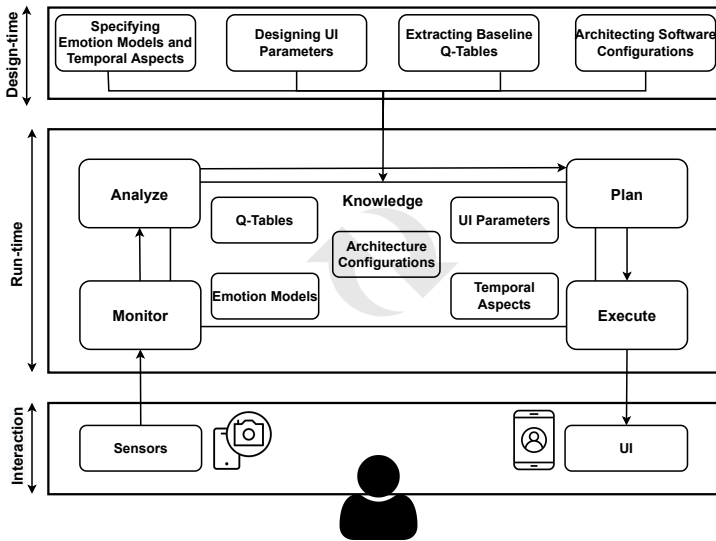


Fig. 3. The emotion-driven adaptation approach. At *design-time* pre-designed UI/architectural models and parameters are adopted and fed to run-time. At *run-time*, the control over the environment is established.

(quantitative) and hermeneutic circle (qualitative) [16], thus, following a mixed data analysis approach.

Understanding various aspects of design, emotions associated with UI parameters, and software qualities from the survey supported the subsequent implementation of the mobile app and experimentation steps. Therefore, the survey data analysis in the *eighth* decision point is followed by a case study in the *sixth* point. Case studies are widely used in software engineering and HCI. Our case study method follows a quantitative process for statistical analysis (*fifth* decision point). Afterward, the *seventh* decision point shows the experimental research followed by assigning participants to experimental conditions in a controlled environment. We performed the experiments in iterations to effectively use the results of each experiment to improve software development. The *eighth* decision point shows that we analyzed the experiment data by statistical analysis and further discussed our research findings.

4 APPROACH

This section presents the main approach for UI adaptation to emotions. Based on the state-of-the-art analysis, we understood that emotion models, their temporal aspects, UI specifications, adaptation baselines, and software architectures should be arranged before run-time adaptation occurs. Such models and considerations at design-time would provide valuable knowledge for the adaptation process and improve its efficiency. As shown in Figure 3, the proposed *Emoticontrol* adaptation approach relies on design-time activities that feed MAPEK-based run-time adaptation processes that interact with the environment.

4.1 Design-time

The design-time step includes the activities performed before running the system to guarantee proper functionality and high quality. As shown in Figure 3, the design-time layer investigates which emotional models should be adopted, what are the time-related aspects of emotions, how the

UI should be designed, how we could support our decision-maker with some insights, and what architectural models provide proper QoS.

In Section 2, we spoke about **adopting emotion models** for classifying emotions. Emotion models play a key role in *Emoticontrol* since they impact the interpretation of emotions based on what humans demonstrate, mainly by their faces. Among dimensional and categorical models, we choose Ekman's basic emotions [21] under categorical models since we require to specify not only the valence but also the exact type of emotions at each time. The second reason for choosing Ekman's basic emotions is that this model is known as a universal emotion model that is not dependent on age, gender, and race [44, 61].

In addition to models, there are **temporal aspects** [1] of emotions to consider in UI adaptation. This dimension emphasizes the application of UI adaptation in a proper frequency or at the right time and consists of two sub-dimensions of *inference* and *application*. The *inference* time could be specified with dimensions such as: *i)* Values from the last few seconds [10], for instance, the inference that happens based on the user's emotions for a certain amount of time (e.g., last 5 seconds). *ii)* From all known values [5], for instance, it can be used to study historical emotion data to identify the typical values of a user's emotions during the interaction. The process above can help to identify the user's emotional peaks. *iii)* The last interaction result computed during the previous use of the system, which is outlined as the inference from the last session [31], for example, the self-declared emotions at the end of an evacuation training session. *iv)* The last event occurrence, for example, the time needed for inferring a change of emotion based on changes in the stimulus.

The application time corresponds to the proper time for applying the adaptation. The inference time corresponds to the needed adequate temporal measurement of a context change. For instance, inference time can be the time required to measure an emotion as a constant emotion (i.e., the emotion that sustains without variation). The most significant dimension of *application* time is frequency, which constrains the maximum number of adaptations. The adaptation frequency limit (the time interval for which an adaptation cycle needs to be executed) can help avoid disturbances due to too frequent changes.

As shown in Figure 3, another design-time component is **designing UI parameters**. In fact, the UI parameters are calculated variables expressing the adaptation that must be made to the UI. For instance, detecting confusion in the user's emotions would lead to increased guidance with a pop-up. In such a case, a UI parameter could have various states, such as visible and hidden. In line with the temporal aspect mentioned above, UI parameters can be associated with an additional character called the maximum adaptation. In the *EvacuationApp* case study, font size, background color, pop-up, and map type, are defined as UI parameters that can be changed from one state to another (See Table 1).

The last dimension of the design-time layer is designing **software architectures** that satisfy QoS requirements. There are three generic architectures of adaptive systems [47, 48, 74] (namely centralized, distributed, and hierarchical) that could be adopted based on dynamic situations and requirements. In the architectures, the MAPEK loop receives inputs and affects the physical layer in which physical devices are located. The architectures are specified based on distribution and collaboration characteristics. This means that data analysis software sometimes needs to be deployed on a single node (*centralized model*) and sometimes on several dispersed nodes (*distributed* and *hierarchical models*). The centralized mode comprises processing on a central local or remote controller. If an application senses emotions and positions data, and processes the data on a single processor, the centralized model is adopted. The distributed model includes processing on independent or collaborative controllers. Thus, if the emotions and location are analyzed on two servers with possible data sharing for coordination, the distributed model is used. The hierarchical

model contains independent or hybrid (i.e., with distributed collaborative) controllers. In our example, emotions could be processed locally on users' phones, and if a more CPU-demanding process is required, that could be performed on, e.g., the cloud. Assessing these architectures and their associated QoS could help engineer a quality-driven architecture at design-time to feed the run-time knowledge-base with valuable input.

4.2 Interaction

Self-adaptive systems constantly interact with their environment [74]. They receive sensory inputs and give adaptation commands to the part of the environment they control. The interaction layer is a part of the environment and includes **sensors**, which are physical or software components, and the screen surface where the **interface** is rendered. In the *Emoticontrol* approach, all sensors provide raw data at run-time that are transferred to the run-time layer. The interface is used for specific goals (such as routing, gaming, and so forth) and is supposed to adapt continuously to the detected contextual factors. For example, in the *EvacuationApp* case study, the central sensor in facial emotion recognition is the built-in mobile camera, and the mobile screen facilitates the rendering of the interface. The participant's facial expression is captured continuously through video feeds using their mobile phone camera. At run-time, these video feeds are sent to FaceAPI, an emotion analysis library, at a rate of ten frames per second (video is a collection of image frames).

4.3 Run-time

The run-time layer handles the adaptations when the software system is running. As previously explained, the process to follow is conceptualized using MAPEK control loop [74] that includes *monitor*, *analyze*, *plan*, and *execute* under a comprehensive *knowledge* base. Those steps are introduced to facilitate adaptation in case of changes and uncertainties. Sensors produce raw data, in our case, emotions from the user's face. In such a process, the **monitor** step filters raw data into insightful data and watches sensors changes and infers values' changes. Afterwards, the **analyze** computes inferred values from raw values. In **plan** step, the adaptation process performs the decision-making by planning the application of the UI adaptation and getting input from the control loop knowledge base. The control loop **knowledge** base contains both the historical data as well as the inputs from the design-time step actions (explained above). Following the plan step, and if the adaptation is not stopped, the **execute** step modifies the interface. In *EvacuationApp*, the camera feeds data into the FaceAPI library, which uses pre-trained models to predict which emotions are likely to be expressed by a user. FaceAPI then passes that information to the rule-based, sequential, or MFRL-based engine. The engine then uses the predefined rules or the MFRL to plan which adaptation(s) should be triggered. When a decision has been made, that info is passed to the adaptation context within execute, which alters its UI state.

The process described above has a high dependency on the adaptation decision-maker, which is the core of the plan step. The role of the **decision maker** is to select the optimal action (change of UI) by taking into account the current emotional state of the user as well as the current UI of the application. To this end, our approach makes use of an RL technique, Q-learning [73]. Q-learning is a widely used method for decision-making scenarios due to its ability to come up with optimal decisions through a model-free learning approach. The key part of using Q-learning is to divide the problem into a set of states, S , actions, A and rewards, R . The state space of our Q-learning approach is determined by two important attributes: *i*) Emotion categories, E and *ii*) Interface configurations, I .

Algorithm 1 Interface Configuration Selection Algorithm

Require: :

```

1: States  $S = \{s_1, s_2, s_3, \dots, s_n\}$ 
2: Actions  $A = \{0, 1, 2, \dots, m\}$ 
3: Labels  $L = \{l_1, l_2, l_3, \dots, l_p\}$ 
4: Rewards  $R = \{r_1, r_2, \dots, r_n\}$ 
5: procedure DECISION-MAKER( $S, A, L, R, I, \alpha, \gamma$ )
    configuration
    6:    $E \leftarrow \text{identify\_category}(I, L)$ 
    7:    $s \leftarrow (E, i)$ 
    8:    $r \leftarrow R[s]$ 
    9:    $(s', a') \leftarrow \text{argmax}_a Q(s, a)$ 
    10:   $Q'(s, a) = (1 - \alpha) * Q(s, a) + \alpha * (r + \gamma * \max(Q(s', a)))$ 
    11:   $a \leftarrow a'$ 
    12:  return  $a$ 
    13: end procedure
  
```

▶ represents the selection of each $i \in I$
 ▶ emotional categories
 ▶ Reward for each of the state
 ▶ Find the state of the system from the emotional category and current interface
 ▶ Get Category from Forecasts
 ▶ combine category and interface configuration to form the state
 ▶ Reward for attaining the state, s
 ▶ The action to reach that state

Definition 4.1. (Emotional Categories) We define Emotional Categories, E as a discrete set $\{e_1, e_2, \dots, e_n\}$ where e_j represents the emotion of the user captured by the user application by mapping the analysis of the facial image obtained to a unique label, $l \in \mathcal{L}$.

Based on this, the state space, $S \subseteq E \times I$ is defined over the set of possible emotional categories and the set of all interface configurations in I .

For instance, let $E = \{\text{joy, surprise, fear, disgust, anger, sad, neutral}\}$ represent the emotional categories and let $I = \{\text{dark, light, 3dmap}\}$ represent a set of interface configurations. Then the state space S consist of all possible combinations of $E \times I$ such that $S = \{\text{joy_dark, joy_3dmap}\dots\}$.

Action space, A , consists of a set of actions $\{a_1, a_2, a_3, \dots, a_m\}$ such that $a_j \in A$ denotes the selection of a configuration s_j in I . Rewards, R on the other hand is a set $\{r_1, r_2, \dots, r_n\}$ where $r_j \in Z$ denotes the reward value for attaining a state, $s_j \in S$.

S and A together form a $n \times m$ matrix where n denotes the number of states and m denotes the number of actions. Q-Table forms the heart of the Q-learning approach where each value corresponds to a (s, a) pair and its value denotes the relevance of selecting an action, a from a state s . In other words, it denotes the benefit of selecting an interface configuration a_j given the current interface configuration in use and the emotion of the user. This property is leveraged by the selection process to select the optimal interface configuration during the process of UI adaptation. The complete algorithm based on Q-learning is presented in algorithm 1. It uses two key parameters, α and γ where $0 < \alpha \leq 1$ denotes the learning rate which represents the importance given to the learned observation at each step, and $0 < \gamma \leq 1$ denotes the discount factor, which can be considered as the weight given to the next action (selection of an interface configuration).

The algorithm first identifies the emotional category from the facial image of the user, E based on the labels, L . The E identified along with the running interface configuration, I is combined to identify the current state, s inside the Q-Table (lines 6-7). The algorithm then assigns a reward for attaining the state, s . Following this, the maximum value of the state-action pair, (s', a') that can be reached from the current state is identified (lines 8-9). The Q-table is then updated using the Q-function (line 10) and the action (instance, $\hat{s}$) is returned (lines 11-12).

In this manner for every query, q , our algorithm selects the optimal interface configuration i by selecting the action that maximizes the reward. By assigning negative rewards (penalties) to a selection of interface configuration that leads to an undesirable emotional category (such as anger), the approach ensures that any incorrect selection is penalized in the form of a high negative reward and this means as the time progresses, the algorithm continuously improves the selection process to ensure that the UI is adapted to maintain an emotional balance of the user by ignoring configurations that can lead to high penalties.

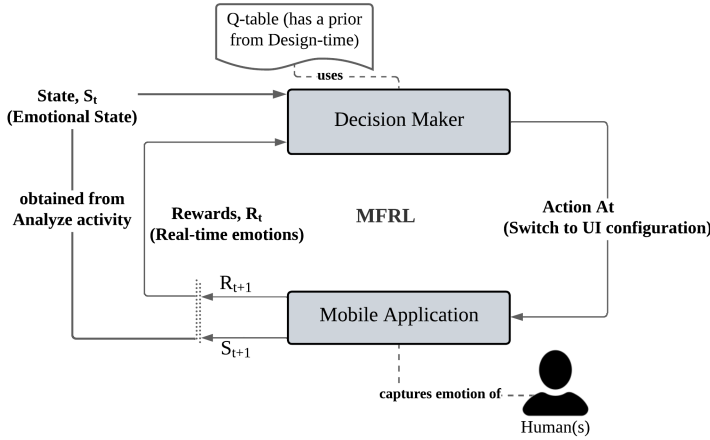


Fig. 4. Working of MFRL Approach

Figure 4 depicts the overall working of the MFRL approach. It reads the emotional state of the users at run-time and uses that as a driver for adaptation. Based on this, the Decision Maker selects an action that corresponds to an adaptation of the user interface. The emotion of the user captured post the adaptation of the UI is used to determine the effect of the adaptation. This is used as a reward/penalty in the subsequent adaptation cycle to improve decision-making. As stated above, in Q-learning, the agent does not require any model of the environment. This translates to the fact that the agent learns about the environment through its interactions and feedback. However, during the learning process, the agent has to perform many actions to measure the impact of the action on the system. This process is known as exploration. On the other hand, the agent also uses the learned knowledge to perform actions. This process is known as exploitation. To balance exploration and exploitation, Q-learning uses an ϵ -greedy exploration strategy, where the agent selects the action with the highest Q-value with a probability of $1 - \epsilon$ and selects a random action with a probability of ϵ . The value of ϵ determines the degree of exploration vs. exploitation. For example, a high value of ϵ will result in more exploration (selecting more random UI adaptations), while a low value of ϵ will result in more exploitation. To counter this, in our approach, we perform an initial user study to understand the relationship between users' emotions and the expected UI configuration. This is then used to create an initial version of the Q-table. Post the analysis, we obtain a mapping, M where for each user $u_i \in U$ we obtain a preference vector, $v = \langle b_1, b_2, b_3 \dots b_n \rangle$ where each $b_i \in B$ is inturn a vector, $\langle c_1, c_2, \dots c_n \rangle$ s.t c_i denotes one of the possible n UI configurations. This vector represents the preferred interface of the user for each emotion. For each $b_i \in B$, a preference value, p , is computed by calculating the number of c_i that is preferred the most a given b_i across all users U . This p value is computed for all the (b_i, c_i) pairs, and these values become the values for (s, a) pairs to form a base Q-table. As depicted in the figure in our scenario, the decision maker uses this Q-table at run-time. This is to leverage the knowledge gained on the run-time to converge towards optimal decisions.

5 SURVEY

Before implementing the *EvacuationApp* and performing experiments, we needed to gain insights into design, development, and emotions' consideration. Thus, in line with guidelines [42, 69], we distributed an online **survey** to our audience. We followed a structured question-and-answer format

with multiple-choice and explanatory questions to give the participants the freedom to discuss their opinions and ideas. We chose this method since emotion-based adaptation is a growing concept, and its characteristics should still be discovered. The four primary aims of conducting the survey were to *i)* obtain insights on the impact of adaptations on emotions, *ii)* formulate the hypotheses, *iii)* extract a first Q-Table for the MFRL adaptation engine, and *iv)* improve the design aspects.

5.1 Insights on Emotions and Adaptations

In summary, we aimed to answer the following three questions.

1) *What type of information would the participants like to see in the application for learning evacuation?*

34% of the participants mentioned that the essential information to show is the evacuation routing, time constraints, and safe evacuation guidelines. 9.6% of the participants found emotional intensity a vital factor for successful evacuation training. These results helped us decide that the main focus should be evacuation routing based on emotions.

2) *What emotions do participants experience during the learning sessions?*

46% of participants have already participated in emergency preparedness sessions (including information-sharing sessions and emergency tests). Among the participants with prior learning session experience, 16% found the experience extremely useful. The reason behind the dissatisfaction among other participants was mentioned as a lack of realism of the session and not emotionally engaging sessions (77%). They found the information provided in a vast scope; therefore, the information was forgotten by the sessions participant. Participants reported that they experienced specific emotions in a successful preparedness session: feeling surprised, joy, and fear. The particular report was in line with prior research on the type of emotions involved in emergencies [49]. These results were helpful for us to set target emotions in our adaptation mechanisms. We designed the app in an engaging and realistic way as well.

3) *What emotions are involved in various types of adaptation in evacuation learning?*

To define the proper type of UI adaptation to emotions, we showed different UI adaptation possibilities to the participants. We showed the specific user interface before adaptation and after adaptation. In the next step, we asked participants to declare their emotions regarding a specific adaptation. The emotions are based on Ekman's six basic emotions model on a 7-point Likert scale. The first adaptation concerns changing the font size aroused respectively neutral ($\bar{x}=4.19$), fear ($\bar{x}=3.36$), and surprise ($\bar{x}=3.25$). The second type of adaptation is adding pop-up information for the learner during evacuation. Participants declared surprise ($\bar{x}=4.69$), anger ($\bar{x}=4.03$), and fear ($\bar{x}=3.97$). The third type of adaptation concerns changes in the UI background color. Participants declared neutral ($\bar{x}=4.05$), surprise ($\bar{x}=3.22$), and joy ($\bar{x}=3.08$). The fourth type of adaptation is changing the map from a street to a satellite view. The participant declared anger ($\bar{x}=4.06$), surprise ($\bar{x}=4.03$), and fear ($\bar{x}=3.78$). The results of declared emotions engrave the groundwork of the experienced emotions in various adaptations. We base our adaptations on the results of this phase.

4) *What level of response time EvacuationApp should provide?*

The main motivation behind this question was to understand from the users how fast the application should respond to plan the adaptation accordingly. This question had multiple answers: less than 100ms, 500ms, 1s, 2s, and 3s. We decided on these options since users in the mobile application domain will abandon interactions or even delete apps if response times exceed 3 seconds [56]. More specifically, if the response time becomes high in safety-critical situations, it endangers people's lives. Most participants who answered this question required a response time of less than 1 second (46%). 22% of the participants even needed a lower response time of lesser than 100 milliseconds, while few participants were comfortable with higher response times of 2 or 3 seconds. The results were helpful and gave us insights regarding the expected level of QoS stated by the potential users.

5.2 Hypotheses

Based on the above-mentioned results of the survey and our literature review, we developed four hypotheses to be validated by experiments. We formulated our hypotheses by following the SMART mnemonic [15, 33, 57]. The SMART acronym stands for Specific, Measurable, Attainable, Relevant, and Testable. In our approach, the goals are Specific that are regulating participants' emotional reactions to the UIs. Measurable since the experiments could show how intense the target emotions are with numbers. Attainable in that it is possible to accomplish with experiments. Relevant in that the adaptation system could perform better by learning such reactions to UIs, and Testable with real numbers on intensity and duration. The hypotheses should be formulated correctly to clearly state the parameters of the experiment and communicate the expected relationship. The experiment data analysis should statistically prove or refuse the hypothesized relationship [25].

- **Hypothesis 1.** participants' joy intensity level is triggered by changes in the font size as a UI parameter.
- **Hypothesis 2.** participants' surprise intensity level is triggered by showing a pop-up dialogue and/or Satellite v.s. Street map view.
- **Hypothesis 3.** participants' fear intensity level is triggered by the background color as a UI parameter.
- **Hypothesis 4.** Adopting a cloud infrastructure when required keeps the recommender system's response time less than a second.

Mackenzie [39] suggests two ways of hypothesis testing: parametric and non-parametric tests. Parametric tests operate on data from a probability distribution, such as the normal distribution or the t -distribution. In contrast, non-parametric tests are distribution-free and make no assumption about the distribution of the underlying data. In this research, we mostly used the analysis of variance (ANOVA), the most widely used parametric procedure. Using ANOVA, we give an F -statistic, the degrees of freedom for the F -statistic, and the associated p -value. In fact, an ANOVA determines if an independent variable (the test conditions) had a significant impact on a dependent variable (the measured responses).

5.3 Extracting a First Q-Table from the Survey

The downside to a Q-learning approach is that it takes some time for the agent to explore all possible actions from all possible states. In the context of our experiment, since we are working with humans, we cannot simply perform 10000 iterations as training, where we put users through all possible states and actions. Thus, the survey helped extract some general rules and constraints. We implemented constraints into the agent by adding values to the Q-table. The agent will use these Q-values to pick one action over other actions from specific states. For instance, assume our agent has taken action that changed the background color of the application to white. In this case, it would almost certainly result in negative emotions from the user if the following action by the agent was to change the text color to white. The reason is that you cannot read white text on a white background. This can be seen as an edge case with a particular state-action pair that the agent should *not* explore. In practice, we would give the action of changing the text color to white a negative Q-value for all situations where the background state is white. This means the agent will never pick the action of changing the text color to white whenever the background color is white. Apart from this obvious example, knowing the target emotions and actions that (based on survey participants' declaration) could positively and negatively impact those target emotions, we set the constraints to facilitate the emotion goals.

5.4 Mobile Application Design

We designed the *EvacuationApp* in various iterations by following the usability for mobile UI [55], and user-centered [32] approaches, as well as the feedback received from the survey participants. We received the following feedback from the survey: *i)* The focus should be on the map service, an essential UI element. *ii)* There should be no video preview of the user or unnecessary buttons. This ensures that the user will not be distracted by the video of their face, and they do not have to manually click a button to start the experiment (so to minimize unnecessary manual interaction). *iii)* There should be no labels explaining the meaning of the markers (like current location and destination) since it should be clear using, for example, icons. *iv)* Users require adequate interaction with the map and have a suitable emotions regarding their physical context. Figure 5 in Implementation Section shows the final UI design. The starting point and the destination are shown by some icons. The ability of dynamic interaction with the map (zoom in and out, rotate, contextual information) is added. Furthermore, the amount of text on the screen is kept minimal. In addition to the main view for emergency evacuation training, the app has some views to receive the users' declarative feedback after each experimentation, where the interaction was kept as easy as possible (see Fig 7).

6 IMPLEMENTATION

The previous section mentioned that our implementation includes several pieces that are integrated into a mobile app. We go through the implementation of adaptation engines as well as the mobile application and the software architecture.

6.1 Implementation of the Adaptation approaches

In order to evaluate the effectiveness and efficiency of our approach for the hypothesis, we defined three different adaptation methods to apply adaptations to *EvacuationApp* user interface. Among the mentioned methods, *Sequential Adaptation* and *Rule-based Adaptation* formed the experimental baselines. *Sequential Adaptation* represents a scenario where the UI keeps changing following a sequential order of changes. *Rule-based Adaptation*, however, represents the approach where the adaptation is performed following a predefined set of rules typically encoded into the application. These rules are defined at design-time to perform adaptations at run-time. We used *Sequential Adaptation* and *Rule-based Adaptation* methods as baselines for comparison of the effectiveness and efficiency of our MFRL approach. MFRL uses a reinforcement learning algorithm to adapt UI dynamically and continuously learn with every adaptation.

We defined two criteria to form the ground basis for comparing the three mentioned approaches. The first criterion consists of the UI elements that are affected by adaptations (See Table 1). More specifically, we incorporated four types of adaptations to training users for safe evacuation (Figure 5). The second criterion concerns the target emotions in applying different UI adaptations. We defined joy, surprise, and fear as the target emotion in UI adaptations in order to improve the evacuation rate. In the following subsections, we first introduce our baseline approaches (i.e., sequential adaptation and rule-based adaptation). In Section 6.1.3, we will explain our novel approach using the MFRL for UI adaptation to emotions.

6.1.1 Sequential Approach. This is our first baseline. It performs predefined random adaptations to a UI element every twenty seconds. We chose twenty-second intervals because emotions take, on average, twenty seconds to reach the highest intensity due to external stimuli [10, 29]. We called this approach *sequential* adaptation. The adaptation sequence is chosen randomly among the following:

- (1) Make the font size larger,

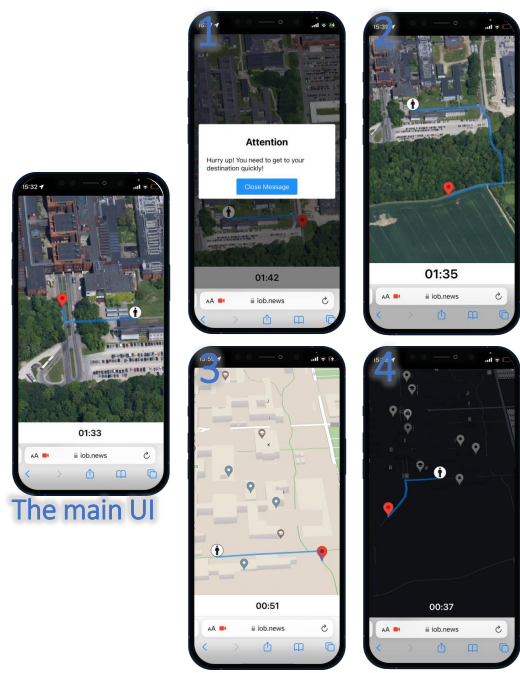


Fig. 5. The main UI and the four types of adaptation: 1. Pop-up information, 2. Font Size, 3. Street View, 4. Background color.

Table 1. The states of UI elements. The state could transit from State 0 to State 1 and vice versa.

UI Element Type	State 0	State 1
FontSize	Medium	Large
ColorTheme	Light	Dark
PopUp	Hidden	Visible
Maptype	SatelliteView	StreetView

- (2) Change the color theme to dark,
- (3) Change the map type from satellite view to terrain view and vice versa, and
- (4) Display a pop-up message.

6.1.2 *Rule-based Approach.* The rule-based method forms the second experimental baseline. In this approach, the adaptation is performed based on a set of predefined rules executed using a MAPE-K loop. The rules are designed based on the hypotheses stated in Section 5.

Each rule has a property that demonstrates its priority compared to other rules. Precisely, the priority property helps select the correct adaptation in case two or more adaptations are triggered simultaneously. To avoid an infinite number of adaptations, we considered a temporal aspect from a UI adaptation problem space [1] and set the maximum adaptations to six in each 2-minute experimentation. As previously mentioned, the defined rules intend to maximize the three specific emotional intensities (i.e., joy, fear, and surprise). Based on the literature and survey studies, we understood that learning requires arousing joy, and simulating danger requires arousing fear and surprise. Table 2 demonstrates the rules and their corresponding descriptions used to implement

Table 2. Table describing the different rules (and their description) that were used for implementing the rule-based engine baseline approach.

Rule	Condition	Description
Priority 1	<i>if (surprise < disgust) or (surprise < sad) → add pop-up information</i>	Goal is to invoke surprise affective state. Display pop-up if the user is disgusted or sad [63]
Priority 2	<i>if (joy < disgust) or (joy < sad) or (joy < anger) → font size</i>	Goal is to invoke joy affective state. Change font size to trigger more joy when user feels sad, anger or disgust [60].
Priority 3	<i>if (fear < 0.08) or (disgust > 0.01) → satellite view to street view</i>	Goal is to invoke fear affective state. Change the map view if the user is not in a fear state or disgust state [8, 59].
Priority 4	<i>if (sadness > fear) → background color</i>	Similar to Rule 3, Goal is to invoke fear affective state. Change the background color if the user is more sad than fearful [58].

this baseline. These rules have been created based on different literature results, as cited in the table. These rules were created to verify the different hypotheses listed in Section 5.2.

6.1.3 Model-Free Reinforcement Learning-based Approach. The MFRL approach represents the main contribution of this work. As stated above, our goal is to compare the performance of this approach (MFRL) with the two baseline approaches presented above (rule-based and sequential). These comparisons further serve as the basis for the hypothesis mentioned in Section 5.2. In this approach, the adaptation is performed using the Q-learning algorithm 1 described in Section 4. We use parameters $\gamma = 0.9$, $\alpha = 0.5$. These values were fixed based on the sample experiments' overall requirements and feedback. We use *TypeScript* for implementing the algorithm. We employ FaceAPI to capture the user's emotion at run-time constantly. The emotion is further mapped into one of the six Ekman basic emotions [20] and neutral. Based on the emotion identified, an integer reward/penalty is assigned. Table 3 presents the values of the integer rewards/penalties assigned by MFRL based on the identified user emotion. These values are decided based on evidence from the literature [43, 73] and the results of our user studies (Section 5). Further, the following objectives were considered for assigning the reward and penalty values:

- Rewarding the agent for reaching the goal state: MFRL assigns a positive reward to the performed adaptation when it reaches the goal state. The most important goal state in our scenario is *joy*; hence we assigned a high value (5) compared to other emotions. This also corresponds to Hypothesis 1. This reward is kept high enough to motivate the approach to find the optimal set of adaptations to attain the goal state.
- Penalize the approach for performing suboptimal adaptations: MFRL assigns a negative reward to the adaptation performed when it selects adaptation strategies (actions) that don't lead to the goal state or lead to an unintended emotion such as Disgust, Anger, or Sad with Disgust being the most unintended emotion. To this end, we assigned the maximum penalty value for the emotion Disgust followed by Anger and Sad. This penalty is kept high enough

Table 3. Reward / Penalties associated with basic emotions.

Emotion Number	Emotion	Reward / Punishment
0	Joy	5
1	Fear	4
2	Surprise	3
3	Neutral	0
4	Sad	-1
5	Anger	-2
6	Disgust	-3

to discourage the agent from taking the adaptations that trigger unintended emotions in the future.

- Use intermediate rewards: Intermediate rewards which are not high and not low are used to incentivize the adaptation approach for taking actions that move it closer to the goal state. Since Fear and Surprise are emotions that are closer to the goal as opposed to other emotions in an emergency scenario, we keep these rewards a little smaller than that of Joy. This further corresponds to hypotheses 2 and 3. These rewards are assigned in such a manner that they are smaller than the final reward for reaching the goal state but high enough to encourage the approach to select those adaptations that lead to intermediate states.
- Balance rewards and penalties: Neutral is an emotion that is neither towards the goal nor too away from it; hence we use it as a balanced state. The approach is neither penalized nor rewarded.

Moreover, these values are also assigned, taking into consideration that the total rewards and penalties assigned to the agent are balanced so that the approach is neither overly incentivized nor overly penalized. It is also important to note that the integer values used for penalty and rewards, in this case, may not be the same while dealing with another case. Since it depends on the context, the target set of emotions, and the dynamics involved. Designing an optimal set of values for rewards/penalties often requires trial and error to find the optimal balance of rewards and penalties for a given problem. In some cases, it may be necessary to use more advanced techniques such as shaping or curriculum learning to help the agent learn the optimal policy.

The value assigned is further leveraged to populate the Q-table as depicted in algorithm 1. We use of a multi-dimensional array for Q-table. This allows us to select the best possible action from any given state (represented by the combination of current user emotion and running UI configuration). Each possible action (such as changing font size, changing color theme, changing dialog box visibility, etc.) is stored in a one-dimensional array. Based on the selected action, the UI is dynamically updated.

6.2 Implementation of Architecture

EvacuationApp is developed as a *React* progressive web application. *React* was selected because it is designed to support highly interactive UIs that change often. The backend is a *NodeJS* application interacting with a *MongoDB* database. The backend is also written in *TypeScript*. The application has integrated several pieces, namely *maps*, *emotion detection software*, and *adaptation engines*. The Map component is implemented using the *Google Maps API*. Specifically, the *Maps JavaScript API* was used to render the map, while the *Directions API* was used to calculate the route between users and their destinations. We used the *FaceAPI* library to infer emotions from the user's face every 100ms and add to the adaptation engine.

The architectural design decisions of *EvacuationApp* could impact various quality attributes [51] such as performance. Several factors may impact performance, such as sampling rate, service times, networking, infrastructure, and computational capabilities (RAM, CPU). While the sampling rate could be adjusted and service time depends on the specification of physical devices, computation, infrastructure, and network depend very much on the position of processing. In our case, the processing could be performed on users' phones, servers, and/or the cloud.

To test the local deployment, we created a React context to collect the data from the React application. Relevant data points collected are: *i*) time of starting face detection, *ii*) time in which the most intense emotion has been picked, *iii*) time in which an adaptation has been completed by setting a value, *iv*) time of the start of the Q-learning, *v*) time of the end of the Q-learning. Based on the results, the rule-based and MFRL have negligible CPU times. That is obvious since all adaptation methods are based on simple lookups. Another aspect is understanding face detection and emotion inference computation time, which was as short as 38.9 ms on average. In total, the results show an average of 66.55 ms system response time (from sensing to adaptation), which is too short to be considered an issue. Looking at the experiments performed locally, we understood that performing the computation on the users' phones is satisfying. The app also scales fine because the algorithms are processed on the phone.

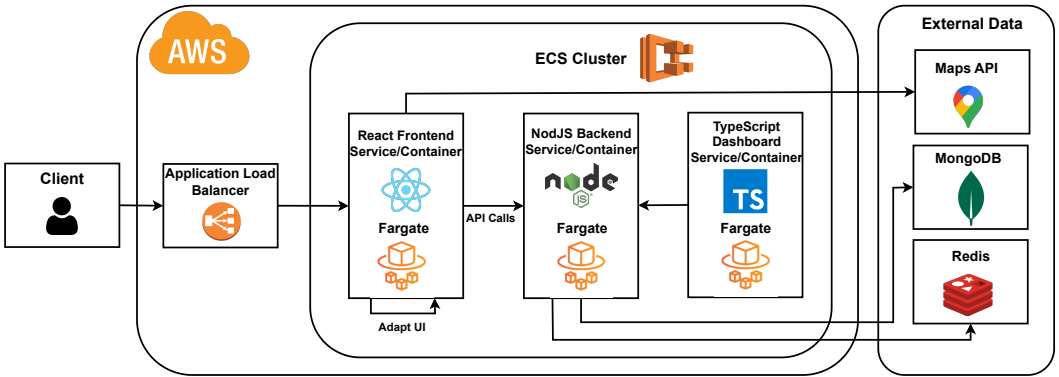


Fig. 6. The cloud-based architecture of *Emoticontrol*.

However, another scalability concern is the backend service, which manages the connection to the database. The backend should get all data regarding users' emotions and task completion, their answers to the questions after each experiment, as well as the admin sign-in and the dashboard webpage. We used the AWS cloud service to avoid QoS issues in the backend service. As shown in Figure 6, we used *load balancing* and *elastic containers* services. A load balancer is used to direct traffic to the different target groups. A target group is the pool of instances provided by the service defined in the ECS cluster. For orchestrating the containers which make up the services (frontend, backend, and admin dashboard)), the Elastic Container Service (ECS) is used. ECS was chosen for auto-scaling, where the number of instances of the virtual server images is increased based on the containers' load (resource utilization). In addition, AWS Fargate for orchestrating the containers within ECS enables the shutting down of containers when they are not used, thus saving deployment costs.

As shown in the figure, to develop the dashboard, we used *JavaScript* and *TypeScript* languages under *React* framework, and the data are stored in *MongoDB*. The dashboard provides insight into the data collected during experiments. We ensured the data was unidentified; for instance, we only

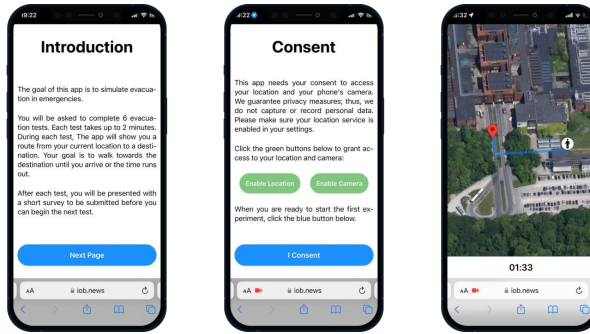


Fig. 7. The experimentation steps.

captured and transmitted the emotions (not videos) and associated it with an ID that could not be tracked. We used Redis to maintain a queue of recorded events, so they would be recovered in case the Backend/DB was busy.

7 EXPERIMENTATION

We followed an iterative approach with two sets of experiments to *i)* improve the *EvacuationApp* based on feedback and observations and *ii)* assess if more MFRL sessions improve the system effectiveness, *iii)* re-approve the validity of the approach. In our iterative development process, the task completion rate is included in the second experiment.

7.1 Experiments Protocol

The experiment is performed through The **EvacuationApp**. As demonstrated in Figure 7, the overall experiment process is as follows. An introduction to the application is first presented to the participant. The introduction consists of the number of experiments a participant will participate in and the duration of the experiment. Next, we request access to the participant camera and location services. In this step, we explain the anonymity of the captured data to participants. Once the location and camera are activated, the participant can click the consent button. After clicking the consent button, the first experiment starts, where participants should follow a path to evacuate themselves. During an evacuation training session, participants may see four types of adaptations already shown in Section 6.1 (Figure 5). For instance, the MFRL adaptation engine may realize that a popup increases an individual's surprise level (a target emotion). So, while the participant follows the map, a popup will notify *hurry up* to arouse a surprise. The system then captures the user's emotion to see if the intended emotion is triggered. Each experiment phase lasts for two minutes maximum (if the participant does not reach the destination). This is done to ensure that the experiment emulates the real evacuation scenario considering the approximately 100 meters distance to destinations and the average walking speed of 0.889-1.508 m/s on a flat surface [4]. After that, the participant is directed to a questionnaire landing page where they provide declarative insights. This process continues four times in the first iteration and six times in the second iteration. In this set of experiments, we used captured emotions, which were derived from participants' facial expressions. It is worth clarifying the ethical approval for safety-critical tests, such as evacuation training. Based on the Denmark research system, ethical consent in such scenarios is not required. However, we considered all ethical, privacy, and safety measures.

Table 4. Emotion arousal and the changes in UI in the first experiment.

Descriptive Statistics				
	Adaptation Name	Mean	Std. Deviation	N
Surprised	ColorThemeDark	0.00165497	0.002109807	22
	FontSizeLarge	0.00347861	0.005692569	22
	MapTypeSatellite	0.00515943	0.013578062	23
	PopupVisible	0.00361154	0.009019287	24
	Total	0.00349761	0.008717492	91
Joy	ColorThemeDark	0.12172483	0.193029685	22
	FontSizeLarge	0.13994357	0.234401062	22
	MapTypeSatellite	0.11450534	0.189850031	23
	PopupVisible	0.11572308	0.183701628	24
	Total	0.12272178	0.19761571	91
fear	ColorThemeDark	0.00418361	0.011965336	22
	FontSizeLarge	0.00106582	0.003187695	22
	MapTypeSatellite	0.00127934	0.00379304	23
	PopupVisible	0.00122646	0.004711659	24
	Total	0.00191591	0.006828711	91

7.2 Experiment 1

The first experiment objective is to determine the effectiveness of the EvacuationApp. Further, we investigate the effectiveness of the MFRL approach to the existing alternative adaptation approaches (i.e., sequential adaptation, rule-based adaptation). The first experiment tests the first three hypotheses (see Subsection 5.2). More precisely, the first experiment looks into the defined type of UI element adaptation to arouse emotions such as joy, fear, and surprise. Twenty-four participants (11 female, 13 male) between 19 to 42 were recruited. This means that the representation of the distribution of gender groups in our research environment was 46% women and 54% men. In most cases, even slight gender imbalance could be among the limitations. While it should be avoided in the future, the MFRL could slightly tackle this since it is generally trained regardless of gender. The experiment duration was approximately 15 minutes. All participants had prior knowledge of using mobile applications. The experiment took place on individuals' cell phones. Participants were provided with a link to access *EvacuationApp*. We asked them to follow the instruction in the *EvacuationApp* and evacuate themselves. The first experiment was divided into four phases. Each phase uses a specific approach to UI adaptation. In sequence, the phases are sequential, MFRL 1, rule-based, and MFRL 2 UI adaptation.

7.2.1 UI Parameters and Emotions (Answer to RQ1). To analyze the validity of the defined UI parameters with our target emotions, we first computed the average emotion intensity aroused by each type of change in UI parameters. The analysis performed in this step is assessed independently of the adaptation method. Table 4 shows the results of the first experiment. In line with our questionnaire, the experiment confirms that joy intensity was the highest triggered by changes in font size. Fear was mainly triggered by theme color change. On average, learners were the most surprised by using the changes in the satellite view. The second type of adaptation that aroused surprise was the pop-up dialogue in line with our literature analysis.

7.2.2 Target Emotions and Adaptation Method (Answer to RQ2.1). To answer the question of MFRL effectiveness in comparison to other approaches in terms of arousing target emotions, we performed a one-way ANOVA test with the target emotions (surprise, joy, fear) as the dependent variable and

the type of experiment (Sequential, MFRL1, rule-based, MFRL2 as an independent variable. This analysis aims to investigate if the independent variable (i.e., type of adaptation) have a measurable effect on our dependent variable.

Our analysis revealed that the type of adaptation had a statistically significant effect on surprise as the target emotions $F(3, 2894) = 10.32, p < 0.01$. It also showed a highly significant effect of type of experiment on fear as the target emotion $F(3, 2894) = 9.626, p < 0.01$. The effect of adaptation type on joy as the target emotion was not statistically significant (see the possible reason in the lessons learned section discussion). In the next step, to understand the primary source of differences in means, we performed a Post-Hoc test using Tukey HSD (Table 5). The result indicated the second iteration of reinforcement learning was significantly more effective in arousing the target emotions (surprise, fear). This result indicates that the MFRL approach correctly improves itself through more iterations.

Table 5. Post-Hoc analysis for target emotions and adaptation method in the first experiment.

Multiple Comparisons					
Tukey HSD					
Dependent Variable	Experiment (I)	Experiment (J)	Mean Difference (I-J)	Std. Error	Sig.
Surprise	RL2nd	Sequential	.004529278438771*	0.000834703	<0.01
		RL1st	.003489306356198*	0.000871724	<0.01
		Rule-based	.002495088155604*	0.000905264	0.03
Fear	RL2nd	Sequential	.001269085063540*	0.000242153	<0.01
		RL1st	.000992190093620*	0.000252893	<0.01
		Rule-based	.000727134669543*	0.000262623	0.029

*. The mean difference is significant at the 0.05 level.

7.3 Experiment 2

We designed the second experiment to achieve several goals. Our first goal was to ensure the validity of the first experiment. The second goal was to assess the effectiveness of MFRL over four iterations. More specifically, to validate the results from our first experiment, we performed four iterations of MFRL approach. The third and final goal of this experiment was to examine our fourth hypothesis. The hypothesis suggests that adopting an edge-cloud infrastructure keeps the recommender system response time less than a second. We recruited twenty-seven participants (12 female, 15 male) between 20 to 58 years old for the experiment. The experiment duration was approximately 20 minutes. The general experimental protocol followed the same steps as the first study. This experiment is split into six phases MFRL 1, No adaptation, MFRL2, Sequential, MFRL3, and rule-based UI adaptation.

7.3.1 UI Parameters and Emotions (Answer to RQ1). We first computed the average emotion intensity aroused by each type of change in UI parameters. The analysis performed in this step is assessed independently of the adaptation method. Table 6 shows the results of the second experiment. The analysis demonstrates a similar result to the first experiments. It confirms that joy intensity was the highest triggered by changes in font size. Fear was mainly triggered by theme color change. On average, learners were the most surprised by using the changes in the satellite view. The second type of adaptation that aroused surprise was the pop-up dialogue which was in line with our literature analysis.

7.3.2 Target Emotions and Adaptation Method (Answer to RQ2.1). Results indicate that type of experiment had a statistically significant effect on all the target emotions with a significance level less than 0.05 (Surprise $F(5, 172166) = 2006.3, p < 0.05$, joy $F(5, 172166) = 1919.23, p < 0.05$, Fear $F(5, 172166) = 2086.18, p < 0.05$). To gain a better insight into the significance level of the

Table 6. Emotion arousal and the changes in UI in the second experiment.

Descriptive Statistics				
Adaptation Name		Mean	Std. Deviation	N
Surprised	ColorThemeDark	0.00018913	0.00018913	22
	FontSizeLarge	0.000239375	0.000411365	23
	MapTypeSatellite	0.00439321	0.011782327	22
	PopupVisible	0.000522933	0.001388599	24
	Total	0.001306236	0.006003879	91
Joy	ColorThemeDark	0.072975863	0.217737655	22
	FontSizeLarge	0.17780563	0.380539433	23
	MapTypeSatellite	0.099668837	0.277902349	22
	PopupVisible	0.011411343	0.024849146	24
	Total	0.089687748	0.261436914	91
fear	ColorThemeDark	0.001794525	0.00733253	22
	FontSizeLarge	0.000122874	0.000250902	23
	MapTypeSatellite	0.000247322	0.001098466	22
	PopupVisible	4.67694E-05	0.000155455	24
	Total	0.000537024	0.003655611	91

different adaptation approaches, we performed a Post-Hoc Tukey HSD. The result shown in Table 7 indicated the third iteration of reinforcement learning was significantly more effective in arousing the target emotions. Moreover, the results indicated that the third iteration of the MFRL approach is performing better than the first and second approaches. This result is in line with our expectation that the MFRL approach refines itself in each iteration.

Table 7. Post-Hoc analysis for target emotions and adaptation method in the second experiment.

Multiple Comparisons						
Dependent Variable		Mean Difference (I-J)		Std. Error	Sig.	95% Confidence Interval Lower Bound Upper Bound
Surprise	Tukey HSD 3rd RL iteration	1st RL iteration	.006204148230151	0.0000914911	<0.5	0.0059434 0.0064649
		No adaptation	.007571266605247	0.0000914911	<0.5	0.0073105 0.0078320
		2nd RL iteration	.003882242690194	0.0000931700	<0.5	0.0036167 0.0041478
		Sequential adaptation	.006804342119558	0.0000914911	<0.5	0.0065436 0.0070651
		Rule-based adaptation	.007245862295789	0.0000914911	<0.5	0.0069851 0.0075066
		Total	.000000000000000	0.0000000000	1.000	0.0000000 0.0000000
Joy	Tukey HSD 3rd RL iteration	1st RL iteration	.061253131394917	0.0021700618	<0.5	0.0550690 0.0674372
		No adaptation	.192931651526901	0.0021700618	<0.5	0.1867475 0.1991158
		2nd RL iteration	.058729068746643	0.0022098828	<0.5	0.0524315 0.0650267
		Sequential adaptation	.023513915947909	0.0021700618	<0.5	0.0173298 0.0296980
		Rule-based adaptation	.085097473183649	0.0021700618	<0.5	0.0789134 0.0912816
		Total	.000000000000000	0.0000000000	1.000	0.0000000 0.0000000
Fear	Tukey HSD 3rd RL iteration	1st RL iteration	.004539045245759	0.0000565092	<0.5	0.0043780 0.0047001
		No adaptation	.002276451255905	0.0000565092	<0.5	0.0021154 0.0024375
		2nd RL iteration	.004541049793804	0.0000575462	<0.5	0.0043771 0.0047050
		Sequential adaptation	.004569287272770	0.0000565092	<0.5	0.0044083 0.0047303
		Rule-based adaptation	.003686157440484	0.0000565092	<0.5	0.0035251 0.0038472
		Total	.000000000000000	0.0000000000	1.000	0.0000000 0.0000000

*. The mean difference is significant at the 0.05 level.

7.3.3 Task Completion and Adaptation Method (Answer to RQ2.2). Our third factor to examine the effectiveness of the MFRL is the completion rate in various types of adaptation approaches. To this end, we performed a one-way ANOVA test, having the completion rate as a dependent variable and the adaptation method (RL1, no adaptation, RL2, sequential, RL3, rule-based) as an independent variable. Our result showed that the adaptation method had a statistically significant effect on the task completion rate $F(5, 166) = 2711, p < 0.05$. In the next step, we are particularly interested to see if the difference between groups of adaptation methods is significant or not. To this end, we performed a Post-Hoc test to examine the last MFRL iteration with the other adaptation approaches as well as the previous MFRL iterations (See Table 8). The between-group analysis revealed a significant difference between the third MFRL iteration and the other types of adaptations.

It showed the third iteration of the MFRL approach outperforms the other adaptation methods. Moreover, the results show that the third iteration of MFRL also improved compared to the first and second MFRL iterations. This is primarily because, in the case of MFRL, continuous learning is happening as opposed to the other adaptation approaches. In the case of a sequential and rule-based adaptation, the adaptation performed is more based on a standard pre-fixed pattern or a choice made depending on pre-configured rules. However, in MFRL, the feedback obtained after each adaptation (through the user emotion captured after every adaptation cycle) continuously improves the adaptation process, which is further enhanced after each iteration.

Table 8. Post-Hoc analysis for task completion rate.

Multiple Comparisons					
Dependent Variable	Experiment (I)	Experiment (J)	Mean Difference (I-J)	Std. Error	Sig.
Completion rate	RL3rd	RL1st	15.11139*	4.142338882	<0.01
		No adaptation	6.85037037	3.249517968	<0.05
		RL2nd	12.69889*	4.142338882	<0.05
		Sequential	8.4644444*	3.249517968	<0.05
		Rule-based	8.22037*	3.293051376	<0.05

*. The mean difference is significant at the 0.05 level.

7.3.4 Target Emotions and Task Completion (Answer to RQ2.3). Finally, we were interested in understanding the correlation between emotions and task completion rate. As shown in Table 9, we measured the correlation between the target emotions and the evacuation completion rate. The results show a positive correlation between the average target emotions and task completion, meaning that arousing joy, surprise, and fear led people to evacuate faster.

Table 9. Correlation analysis between emotions and task completion rate.

Correlations				
Variable		Correlation	Statistic	
			Count	Lower C.I. Upper C.I.
Completion rate	Surprise	0.149	27	-0.245 0.500
	Joy	0.131	27	-0.262 0.487
	Fear	0.109	27	-0.282 0.470

Missing value handling: PAIRWISE, EXCLUDE. C.I. Level: 95.0

7.4 QoS concerns (Answer to RQ3)

Performance is an essential QoS that has a direct impact on usability. In Section 6, we explained our cloud-based architecture and its components. We explained how the architecture uses AWS autoscaling and load balancing. The configuration of auto-scaling is performed on a per-service basis. The decision to scale is based on the average CPU load registered by ECS. Scaling occurs when CPU usage is $> 80\%$. The metrics collected in CloudWatch are done in 1-minute intervals. This means that the scaling latency will be a minimum of a 1-minute interval *plus* the time to spin up a new instance. To be able to scale quicker, the scale-in period is set to 45 seconds. For application load balancing, listeners are set up for different ports, e.g., forcing HTTPS, forwarding dashboard traffic to the dashboard group, and forwarding the rest of traffic to the backend API.

We have created some performance tests based on querying the backend and getting it to scale to > 1 instances of the backend. We set a failure to log in deliberately, to stress the server. We observed no dropped connections, even when the server was pinned at 100%. The service is somewhat slow to scale up, as the decision is based on a rolling average, but the scaling action does happen in due

time. We also observed that when the usage is low, the desired target server count is scaled down. Figure 8 shows that when all four instances are scaled to 2000, the response times remain within an acceptable level. For instance, when the system gets users' data, it experiences the highest response time of 2 seconds for the 50 percentile. For getting all survey data, the response time of 1 second for the 90 percentile is recorded.

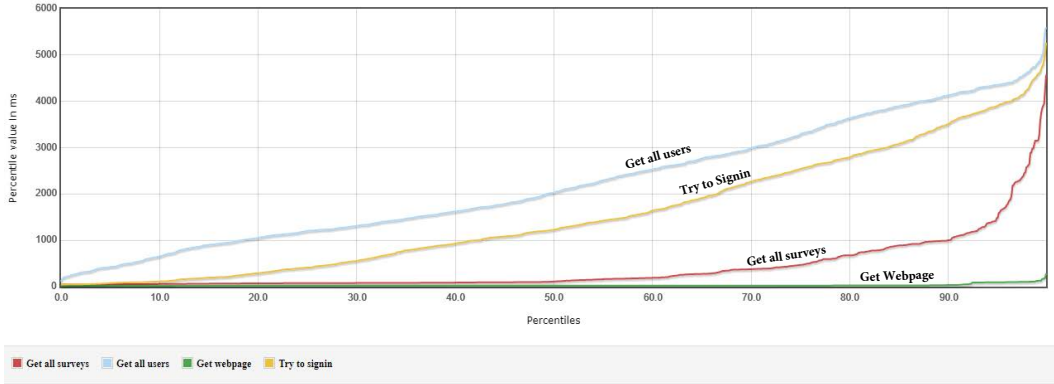


Fig. 8. Percentile backend response time values in ms.

8 LESSONS LEARNED

Putting humans and their emotions as a primary concern of adaptive systems enhances users' quality of experience tied to the system quality of service. This research developed the *Emoticontrol* adaptation approach, an MFRL-based UI adaptation engine to emotions. The general approach is divided into design-time and run-time layers, aiming to anticipate the dynamics before running the system and further adapting the choices based on run-time events. By implementing a mobile application, we performed experiments to answer the research questions on *i*) how UI parameters could arouse specific emotions, *ii*) how MFRL has better effectiveness compared to traditional approaches in arousing target emotions and task completion, *iii*) how the system QoS could be improved by adopting adequate software architectures. We performed 258 experiments (with 51 participants), but more experiments are desirable to confirm the effectiveness of the MFRL approach in UI adaptation. In fact, the presented set of experiments is to explore the approach, gather insights, and pave the way for additional research and practice by the community. While some settings are specific for the EvacuationApp (timings, UI parameters), the *Emoticontrol* approach could be reused in interactive systems to adapt the UIs/content efficiently. To reuse the adaptation system, the states and actions of the MFRL should only be modified to the use case goals.

In the process of developing the *Emoticontrol* adaptation approach and then experimenting using the case study, we learned several lessons:

- Humans have different goals, desires, characteristics, and previous experiences. Thus, creating a general model for human behaviors is difficult. In addition, individuals may show various behaviors to a single stimulus if they are in different physical or social contexts. This was validated in our experiments, verifying the four hypotheses mentioned in Section 5.2. In our experiment, we understood that the MFRL agent improves in iterations by learning and makes better adaptation decisions. This was further demonstrated by the results presented to answer RQ2.
- At the beginning of our research, when we conducted pilot experiments with our colleagues, we started by uploading the videos to be processed on a server. This raised concerns regarding

performance and privacy (see Section 9). We improved our architecture to gather only necessary data, which significantly improved our performance, as hypothesis 4 mentions and the results answering to RQ3 show. We report our privacy measures in future work.

- Along with automatic detection, collecting insights from users by self-reporting methods gives a better picture of the system under development. We gathered users' feedback after each experiment and in each iteration of experiments and improved our design, development, and experiments process. We also created a first Q-Table for our MFRL based on declarative emotions associated with the UIs. This enhanced the performance of our adaptation approach investigated by RQ2. This was demonstrated in the results in Table 5 and Table 7 where we can clearly see how having an initial Q-table enhanced the adaptation effectiveness. Thus, depending on the nature of the case, we suggest adopting an iterative approach with humans in the loop and enhancing the interaction and UI in iterations.
- In the experiments, we noticed that people get frustrated with too many adaptations, which may negatively impact their mood and performance (which would be against what RQ2 aims for). For instance, adopting a rule-based approach that does not consider temporal aspects to, e.g., set the maximum number of adaptations, does not really work in practice. Using our MFRL solves this problem, but if still using the traditional adaptation approaches is preferred in cases, then considering the temporal aspects of adaptation should be prioritized.
- We understood that MFRL is quick to pick an adaptation; thanks to the combination of edge and cloud infrastructure, it imposes no performance issues, further validating our hypothesis 4. While the selected algorithm does not affect performance, the hardware of the client and React itself play a huge part. Better performance might be achieved with a native app or a language that does not rely on the browser event loop as JavaScript does. However, this will most likely be the case for many front-end frameworks.
- We understood that arousing emotions could be supported by other actuation (not only UI changes) in the environment. A more comprehensive (and complex) model in future work would consider dynamic environments (such as light and temperature) as the states of the adaptation model (in addition to emotions and UI parameters) and use, e.g., an IoT system actuators to impact users' emotions and consequently behaviors.

9 THREATS TO VALIDITY

Threats to *construct validity* are related to the use of a controlled experimental setup, the choice of participants, the privacy of experiments, choice of time interval and incorrect selections of adaptations (arising from the use of a model-free learning approach which may put the users in distress). To address this issue with the choice of participants, we recruited participants keeping a clear eye on the diversity. We agree there was a slight imbalance, but this will be addressed in our future work. We also ensured that all participants had prior knowledge of mobile phones. To address the issue associated with privacy, we ensured that neither our study nor the approach itself stores the identity of the individual. For instance, at run-time, we store the ID dynamically assigned to the app rather than the names or details of the individual. A time interval of 2 minutes was used in the experiments to ensure that the given task emulates a real evacuation scenario as closely as possible where a human has to quickly follow the instructions to get to the destination. Moreover, our approach does not depend on the time interval but rather the emotion to adapt the UI, hence we believe that this can be extended to any time interval values. Further, to address the issue due to incorrect adaptations, the initial study resulted in the formation of a base Q-table which was further used to perform individual adaptations at run-time. In addition, multiple experiments were also performed to validate our hypothesis.

Threats to *internal validity* concern using only a basic set of emotions. Even though this is the case, as we have explained in Section 6, we have tried to use only well-defined basic emotions. More work is required to enhance the approach to handling the uncertainty induced by complex emotional evolution and emotion combinations. Furthermore, capturing emotions from only the face may cause an accuracy challenge. Using multimodal emotion detection approaches to feed our *Emoticontrol* MFRL engine can help achieve better results. Moreover, further work is required to consider different external factors (other than the UI) that would impact the user's emotions when making an adaptation decision, such as environmental context.

Threats to *external validity* concern the scalability, generalizability and ecological validity of our approach. Although the approach has been evaluated on a well-defined case study, further experimentation is needed to test and evaluate its scalability (e.g., using Deep Q-Learning [72]) to solve the effectiveness and efficiency issues that might arise from the increased state and action space. Another threat to external validity could be the controlled environment and the possibility of experimenting in real cases. Since this is a training application, the app is also used in a real environment without any risk. Concerning the generalizability, even though our approach has been tested only using six emotions and one application, the approach can be extended to support more emotions and different applications. However, further experimentation is required to evaluate this capability. Regarding ecological validity of our approach, since the study was conducted with real participants in an actual environment with a scenario that emulates real-life settings as closely as possible, we believe that our study results can be generalized to different real world scenarios. However, further experiments with different scenarios needed to demonstrate the same.

10 CONCLUSION

This paper presented *Emoticontrol*, an emotional UI adaptation approach based on model-free reinforcement learning. The approach is conceptualized based on the MAPEK control loop concept, where an adaptation supervisor interacts with the users and their devices to change the UIs and improve the users' emotions and task completion. We compared the MFRL with traditional adaptation approaches such as sequential and rule-based. We applied the approach in a case study where the users used a mobile application that trains users on safe evacuation from a danger zone. The results demonstrated that various UIs could cause specific emotions in users. We proved that the MFRL-based adaptation approach functions better in completing evacuation tasks and arousing target emotions. We also showed that our adaptation system works properly regarding QoS. We believe that intelligent UI adaptation to human context becomes the future of adaptive systems. Such an approach will pave the way forward for UI self-configuration and computation systems' self-adaptation in a trade-off. In future work, we will apply our adaptation engine to more domains, such as human-robot interaction and social robots.

Acknowledgements

This work is supported by the Innovation Fund Denmark for the project DIREC (9142-00001B).

REFERENCES

- [1] Mina Alipour, Sophie Dupuy-Chessa, and Eric Céret. 2021. An emotion-oriented problem space for ui adaptation: From a literature review to a conceptual framework. In *2021 9th International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE, 1–8.
- [2] Mina Alipour, Sophie Dupuy-Chessa, and Eline Jongmans. 2020. Disaster mitigation using interface adaptation to emotions: a targeted literature review. In *10th International Conference on the Internet of Things Companion*. 1–15.
- [3] Reem Alnanih, Olga Ormandjieva, and Thiruvengadam Radhakrishnan. 2013. Context-based and rule-based adaptation of mobile user interfaces in mHealth. *Procedia Computer Science* 21 (2013), 390–397.

- [4] Claudio Arbib, Davide Arcelli, Julie Dugdale, Mahyar T Moghaddam, and Henry Muccini. 2019. Real-time emergency response through performant IoT architectures. In *International Conference on Information Systems for Crisis Response and Management (ISCRAM)*.
- [5] Wendy S Ark, D Christopher Dryer, and Davia J Lu. 1999. The Emotion Mouse.. In *HCI (1)*. 818–823.
- [6] Simonetta Balsamo, Antiniscia Di Marco, Paola Inverardi, and Marta Simeoni. 2004. Model-based performance prediction in software development: A survey. *IEEE Transactions on Software Engineering* 30, 5 (2004), 295–310.
- [7] Matthew L Bolton, Ellen J Bass, and Radu I Siminiceanu. 2013. Using formal verification to evaluate human-automation interaction: A review. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 43, 3 (2013), 488–503.
- [8] Katharine L Bowen, Joanne E Morgan, Simon C Moore, and Stephanie HM van Goozen. 2014. Young offenders' emotion recognition dysfunction across emotion intensities: Explaining variation using psychopathic traits, conduct disorder and offense severity. *Journal of psychopathology and behavioral assessment* 36, 1 (2014), 60–73.
- [9] Margaret M Bradley and Peter J Lang. 1994. Measuring emotion: the self-assessment manikin and the semantic differential. *Journal of behavior therapy and experimental psychiatry* 25, 1 (1994), 49–59.
- [10] Keith W Brawner and Benjamin S Goldberg. 2012. Real-time monitoring of ecg and gsr signals during computer-based training. In *International Conference on Intelligent Tutoring Systems*. Springer, 72–77.
- [11] Tobias Brosch, David Sander, Gilles Pourtois, and Klaus R Scherer. 2008. Beyond fear: Rapid spatial orienting toward positive emotional stimuli. *Psychological science* 19, 4 (2008), 362–370.
- [12] Rafael A Calvo and Sidney D'Mello. 2010. Affect detection: An interdisciplinary review of models, methods, and their applications. *IEEE Transactions on affective computing* 1, 1 (2010), 18–37.
- [13] Antonio Camurri, Gualtiero Volpe, Giovanni De Poli, and Marc Leman. 2005. Communicating expressiveness and affect in multimodal interactive systems. *Ieee Multimedia* 12, 1 (2005), 43–53.
- [14] Nick Cawthon and Andrew Vande Moere. 2007. The effect of aesthetic on the usability of data visualization. In *2007 11th International Conference Information Visualization (IV'07)*. IEEE, 637–648.
- [15] Trevor Day and Paul Tosey. 2011. Beyond SMART? A new framework for goal setting. *Curriculum Journal* 22, 4 (2011), 515–534.
- [16] Jonas Debesay, Dagfinn Nâden, and Åshild Slettebø. 2008. How do we close the hermeneutic circle? A Gadamerian approach to justification in interpretation in qualitative studies. *Nursing Inquiry* 15, 1 (2008), 57–66.
- [17] Hartmut Dieterich, Uwe Malinowski, Thomas Kühme, and Matthias Schneider-Hufschmidt. 1993. State of the art in adaptive user interfaces. *Human factors in information technology* 10 (1993), 13–13.
- [18] Julie Dugdale, Mahyar T Moghaddam, and Henry Muccini. 2020. Human behaviour centered design: developing a software system for cultural heritage. In *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering: Software Engineering in Society*. 85–94.
- [19] Paul Ekman. 1993. Facial expression and emotion. *American psychologist* 48, 4 (1993), 384.
- [20] Paul Ekman. 1999. Basic emotions. *Handbook of cognition and emotion* 98, 45–60 (1999), 16.
- [21] Paul Ekman, Wallace V Friesen, Maureen O'sullivan, Anthony Chan, Irene Diacoyanni-Tarlatzis, Karl Heider, Rainer Krause, William Ayhan LeCompte, Tom Pitcairn, Pio E Ricci-Bitti, et al. 1987. Universals and cultural differences in the judgments of facial expressions of emotion. *Journal of personality and social psychology* 53, 4 (1987), 712.
- [22] Sergio Firmenich, Alejandra Garrido, Fabio Paternò, and Gustavo Rossi. 2019. User interface adaptation for accessibility. In *Web Accessibility*. Springer, 547–568.
- [23] Julián Andrés Galindo, Sophie Dupuy-Chessa, Nadine Mandran, and Eric Céret. 2018. Using user emotions to trigger UI adaptation. In *2018 12th International Conference on Research Challenges in Information Science (RCIS)*. IEEE, 1–11.
- [24] Jose Maria Garcia-Garcia, Victor MR Penichet, and Maria D Lozano. 2017. Emotion detection: a technology review. In *Proceedings of the XVIII international conference on human computer interaction*. 1–8.
- [25] Darren Gergle and Desney S Tan. 2014. Experimental research in HCI. *Ways of Knowing in HCI* (2014), 191–227.
- [26] Jan Gläscher, Nathaniel Daw, Peter Dayan, and John P O'Doherty. 2010. States versus rewards: dissociable neural prediction error signals underlying model-based and model-free reinforcement learning. *Neuron* 66, 4 (2010), 585–595.
- [27] Donald Glowinski, Antonio Camurri, Gualtiero Volpe, Nele Dael, and Klaus Scherer. 2008. Technique for automatic emotion recognition by body gesture analysis. In *2008 IEEE Computer society conference on computer vision and pattern recognition workshops*. IEEE, 1–6.
- [28] Jan L Guynes. 1988. Impact of system response time on state anxiety. *Commun. ACM* 31, 3 (1988), 342–347.
- [29] Brad Hardin and Dave McCool. 2015. *BIM and construction management: proven tools, methods, and workflows*. John Wiley & Sons.
- [30] Sabina Hunziker, Laura Laschinger, Simone Portmann-Schwarz, Norbert K Semmer, Franziska Tschan, and Stephan Marsch. 2011. Perceived stress and team performance during a simulated resuscitation. *Intensive care medicine* 37, 9 (2011), 1473–1479.
- [31] Petar Jerčić, Philipp J Astor, Marc Adam, Olle Hilborn, Kristina Schaff, Craig Lindley, Charlotte Sennersten, and Jeanette Eriksson. 2012. A serious game using physiological interfaces for emotion regulation training in the context

- of financial decision-making. In *20th European Conference on Information Systems (ECIS 2012), Barcelona*. AIS Electronic Library (AISeL), 1–14.
- [32] Eeva Kangas and Timo Kinnunen. 2005. Applying user-centered design to mobile application development. *Commun. ACM* 48, 7 (2005), 55–59.
- [33] Yanal Mahmoud Kilani. 2021. SMART Business Role in Supporting Marketing Strategies among Telecommunication Organizations In Jordan. *Turkish Journal of Computer and Mathematics Education (TURCOMAT)* 12, 6 (2021), 4568–4582.
- [34] Paul R Kleinginna and Anne M Kleinginna. 1981. A categorized list of emotion definitions, with suggestions for a consensual definition. *Motivation and emotion* 5, 4 (1981), 345–379.
- [35] Amit Konar and Aruna Chakraborty. 2015. *Emotion recognition: A pattern analysis approach*. John Wiley & Sons.
- [36] Wouter Kool, Fiery A Cushman, and Samuel J Gershman. 2016. When does model-based control pay off? *PLoS computational biology* 12, 8 (2016), e1005090.
- [37] Baihan Lin, Guillermo Cecchi, Djallel Bouneffouf, Jenna Reinen, and Irina Rish. 2019. A story of two streams: Reinforcement learning models from human behavior and neuropsychiatry. *arXiv preprint arXiv:1906.11286* (2019).
- [38] David Lindlbauer, Anna Maria Feit, and Otmar Hilliges. 2019. Context-aware online adaptation of mixed reality interfaces. In *Proceedings of the 32nd annual ACM symposium on user interface software and technology*. 147–160.
- [39] I Scott MacKenzie. 2012. Human-computer interaction: An empirical research perspective. (2012).
- [40] Tiago V Maia and Michael J Frank. 2011. From reinforcement learning models to psychiatric and neurological disorders. *Nature neuroscience* 14, 2 (2011), 154–162.
- [41] Christian Märtin, Sanim Rashid, and Christian Herdin. 2016. Designing responsive interactive applications by emotion-tracking and pattern-based dynamic user interface adaptation. In *International Conference on Human-Computer Interaction*. Springer, 28–36.
- [42] Elizabeth Martin. 2006. Survey questionnaire construction. *Survey methodology* 13 (2006), 2006.
- [43] Laëtitia Matignon, Guillaume J Laurent, and Nadine Le Fort-Piat. 2006. Reward function and initial values: Better choices for accelerated goal-directed reinforcement learning. In *Artificial Neural Networks-ICANN 2006: 16th International Conference, Athens, Greece, September 10-14, 2006. Proceedings, Part I* 16. Springer, 840–849.
- [44] Sascha Meudt, Miriam Schmidt-Wack, Frank Honold, Felix Schüssel, Michael Weber, Friedhelm Schwenker, and Günther Palm. 2016. Going further in affective computing: how emotion recognition can improve adaptive user interaction. In *Toward Robotic Socially Believable Behaving Systems-Volume I*. Springer, 73–103.
- [45] Nesrine Mezhoudi. 2013. User interface adaptation based on user feedback and machine learning. In *Proceedings of the companion publication of the 2013 international conference on intelligent user interfaces companion*. 25–28.
- [46] Mahyar T Moghaddam, Henry Muccini, Julie Dugdale, and Mikkel Baun Kjærgaard. 2022. Designing Internet of Behaviors Systems. In *2022 IEEE 19th International Conference on Software Architecture (ICSA)*. IEEE, 124–134.
- [47] Mahyar Tourchi Moghaddam, Eric Rutten, and Guillaume Giraud. 2022. Hierarchical control for self-adaptive iot systems a constraint programming-based adaptation approach. In *HICSS 2022-Hawaii International Conference on System Sciences*. 1–10.
- [48] Mahyar T Moghaddam, Eric Rutten, Philippe Lalanda, and Guillaume Giraud. 2020. Ias: an iot architectural self-adaptation framework. In *Software Architecture: 14th European Conference, ECSA 2020, L'Aquila, Italy, September 14–18, 2020, Proceedings* 14. Springer, 333–351.
- [49] Patricia Moravec, Lu Lucy Yan, and Alfonso Pedraza Martinez. 2021. Wildfire Response Operations: Intentional Fear Reduction through Social Media Updates. *Kelley School of Business Research Paper* 2021-05 (2021).
- [50] Vivian Genaro Motti and Jean Vanderdonckt. 2013. A computational framework for context-aware adaptation of user interfaces. In *IEEE 7th International Conference on Research Challenges in Information Science (RCIS)*. IEEE, 1–12.
- [51] Henry Muccini and Mahyar Tourchi Moghaddam. 2018. Iot architectural styles. In *European Conference on Software Architecture*. Springer, 68–85.
- [52] Henry Muccini, Romina Spalazzese, Mahyar T Moghaddam, and Mohammad Sharaf. 2018. Self-adaptive IoT architectures: An emergency handling case study. In *Proceedings of the 12th European Conference on Software Architecture: Companion Proceedings*. 1–6.
- [53] Jonathan Posner, James A Russell, and Bradley S Peterson. 2005. The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and psychopathology* 17, 3 (2005), 715–734.
- [54] Roger Pressman and Bruce Maxim. 2019. *ISE Software Engineering: A Practitioner's Approach* (9 ed.). McGraw-Hill Education, Columbus, OH.
- [55] Lumpapun Punchoojit and Nuttanont Hongwarittorn. 2017. Usability studies on mobile user interface design patterns: a systematic literature review. *Advances in Human-Computer Interaction* 2017 (2017).
- [56] Murali Ramanujam, Harsha V Madhyastha, and Ravi Netravali. 2021. Marauder: synergized caching and prefetching for low-risk mobile app acceleration. In *Proceedings of the 19th Annual International Conference on Mobile Systems, Applications, and Services*. 350–362.

- [57] Robert S Rubin. 2002. Will the real SMART goals please stand up. *The Industrial-Organizational Psychologist* 39, 4 (2002), 26–27.
- [58] Novi Sad, Serbia Gunilla Carlsson, Giedre Gefenaite, Bjorn Slaug, Steven M Schmidt, and Enrico Ronchi. 2020. A review on the role of functional limitations on evacuation performance using the International Classification of Functioning, Disability and Health. (2020).
- [59] Shilpa S Samant, Matthew J Chapko, and Han-Seok Seo. 2017. Predicting consumer liking and preference based on emotional responses and sensory perception: A study with basic taste solutions. *Food research international* 100 (2017), 325–334.
- [60] J Alfredo Sánchez, Ingrid Kirschning, Juan Carlos Palacio, and Yulia Ostróvskaya. 2005. Towards mood-oriented interfaces for synchronous interaction. In *Proceedings of the 2005 Latin American conference on Human-computer interaction*. 1–7.
- [61] Arman Savran, Koray Ciftci, Guillaume Chanel, Javier Mota, Luong Hong Viet, Blent Sankur, Lale Akarun, Alice Caplier, and Michele Rombaut. 2006. Emotion detection in the loop from brain signals and facial images. (2006).
- [62] Annett Schirmer and Ralph Adolphs. 2017. Emotion perception from face, voice, and touch: comparisons and convergence. *Trends in cognitive sciences* 21, 3 (2017), 216–228.
- [63] Andrew Sears and Julie A Jacko. 2009. *Human-computer interaction: design issues, solutions, and applications*. CRC Press.
- [64] Hanan Shteingart and Yonatan Loewenstein. 2014. Reinforcement learning and human behavior. *Current Opinion in Neurobiology* 25 (2014), 93–98.
- [65] José Fernando Silva, João Emílio Almeida, Rosaldo JF Rossetti, and António Leça Coelho. 2013. A serious game for EVAcuation training. In *2013 IEEE 2nd International Conference on Serious Games and Applications for Health (SeGAH)*. IEEE, 1–6.
- [66] Connie U Smith and Lloyd G Williams. 2003. Software performance engineering. In *UML for Real*. Springer, 343–365.
- [67] Jean-Sébastien Sottet, Gaëlle Calvary, Jean-Marie Favre, Joëlle Coutaz, Alexandre Demeure, and Lionel Balme. 2006. Towards Model Driven Engineering of Plastic User Interfaces. In *Satellite Events at the MoDELS 2005 Conference*, Jean-Michel Bruel (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 191–200.
- [68] Jean-Sébastien Sottet, Vincent Ganneau, Gaëlle Calvary, Joëlle Coutaz, Alexandre Demeure, Jean-Marie Favre, and Rachel Demumieux. 2007. Model-driven adaptation for plastic user interfaces. In *IFIP Conference on Human-Computer Interaction*. Springer, 397–410.
- [69] Hamed Taherdoost. 2016. How to design and create an effective survey/questionnaire; A step by step guide. *International Journal of Academic Research in Management (IJARM)* 5, 4 (2016), 37–41.
- [70] Dharmesh Thakkar, Ahmed E Hassan, Gilbert Hamann, and Parminder Flora. 2008. A framework for measurement based performance modeling. In *Proceedings of the 7th International Workshop on Software and Performance*. 55–66.
- [71] Kashyap Todi, Gilles Bailly, Luis Leiva, and Antti Oulasvirta. 2021. Adapting user interfaces with model-based reinforcement learning. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [72] Tanvi VERMA. 2020. Scalable multi-agent reinforcement learning for aggregation systems. (2020).
- [73] Christopher JCH Watkins and Peter Dayan. 1992. Q-learning. *Machine learning* 8, 3-4 (1992), 279–292.
- [74] Danny Weyns, Bradley Schmerl, Vincenzo Grassi, Sam Malek, Raffaella Mirandola, Christian Prehofer, Jochen Wuttke, Jesper Andersson, Holger Giese, and Karl M Göschka. 2013. On patterns for decentralized control in self-adaptive systems. In *Software Engineering for Self-Adaptive Systems II*. Springer, 76–107.
- [75] Timothy D Wilson and Daniel T Gilbert. 2003. Affective forecasting. (2003).
- [76] Claes Wohlin and Aybüke Aurum. 2015. Towards a decision-making structure for selecting a research design in empirical software engineering. *Empirical Software Engineering* 20 (2015), 1427–1455.
- [77] Murray Woodside, Greg Franks, and Dorina C Petriu. 2007. The future of software performance engineering. In *Future of Software Engineering (FOSE’07)*. IEEE, 171–187.
- [78] Eldad Yechiam, Jerome R Busemeyer, Julie C Stout, and Antoine Bechara. 2005. Using cognitive models to map relations between neuropsychological disorders and human decision-making deficits. *Psychological science* 16, 12 (2005), 973–978.
- [79] Enes Yigitbas, André Hottung, Sebastian Mansfield Rojas, Anthony Anjorin, Stefan Sauer, and Gregor Engels. 2019. Context-and data-driven satisfaction analysis of user interface adaptations based on instant user feedback. *Proceedings of the ACM on Human-Computer Interaction* 3, EICS (2019), 1–20.
- [80] Mao Zheng, Qian Xu, and Hao Fan. 2016. Modeling The Adaption Rule in Context-aware Systems. *arXiv preprint arXiv:1609.01614* (2016).

Received October 2022; revised February 2023; accepted April 2023